

GENERAL ORGANIC CHEMISTRY-II

CONTENTS

Topic	Page No.
Theory	01 – 17
Exercise - 1	18 – 27
Part - I : Subjective Questions	
Part - II : Only One Option Correct Type	
Part - III : Match the Columns	
Exercise - 2	27 – 39
Part - I : Only One Option Correct Type	
Part - II : Single or Double Digit Integer Type Questions	
Part - III : One or More Than One Options Correct Type	
Part – IV : Comprehensions	
Exercise - 3	39 – 43
Part - I : JEE(ADVANCED) / IIT-JEE Problems (Previous Years)	
Part - II : JEE(MAIN) / AIEEE Problems (Previous Years)	
Answer Key	44 – 46
Additional Problems for Self Practice (APSP)	47 – 59
Part - I : PRACTICE TEST-1 (IIT-JEE (MAIN Pattern))	
Part - II : NATIONAL STANDARD EXAMINATION IN CHEMISTRY (NSEC) STAGE-I	
Part – III : PRACTICE TEST-2 (IIT-JEE (ADVANCED Pattern))	
APSP Answers	60
APSP Solutions	60 – 61

JEE (Advanced) Syllabus

Fission of covalent bond: free radicals, electrophiles, nucleophiles, carbocations, carbonanions and tautomerism. Acidic and basic strength of organic compounds.

JEE(MAIN) SYLLABUS

Bases, Acids & Tautomerism : Inductive, Resonance and Hyperconjugation Effect, Applications of Electronic Effects, Tautomerism and Acid-base.



General Organic Chemistry-II

Applications of electronic effects:

Th1: Reaction intermediates

Reactive, shortlived, high energy, unstable species, those are formed in the course of organic reactions are called reaction intermediates.

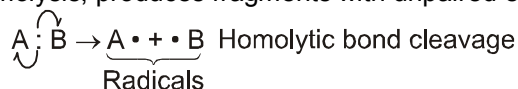
Reaction intermediates generally formed after bond breaking and before bond formation.

A covalent bond can get cleaved by:

- (i) Heterolytic cleavage (ii) Homolytic cleavage.

Homolytic fission of covalent bonds:

The bond may break in such a way that each fragment takes away one of the electrons of the bond. This process is called homolysis, produces fragments with unpaired electron called radicals.

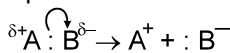


Heterolytic fission of covalent bonds:

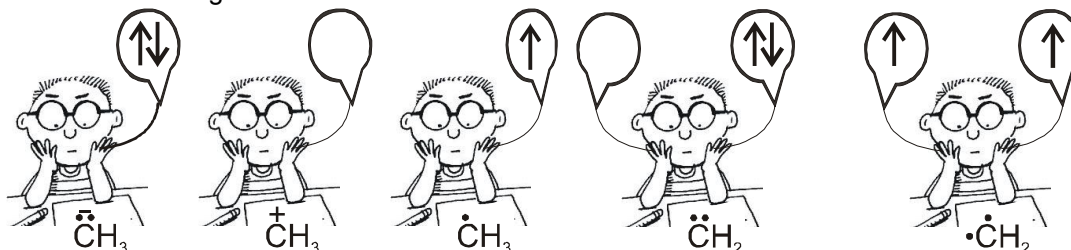
The bond breaks in such a way that one fragment takes away both electrons of the bond, leaving the other fragment with an empty orbital. This kind of cleavage called heterolysis, produces charged fragments or ions.



Heterolysis of a bond normally requires polarized bond.



Polarisation of a bond usually result from different electronegatives of the atoms joined by the bond. The greater the difference in electronegativity, the greater the polarisation. In the given instance, atom B is more electronegative than A.



Section (A) : Carbanions

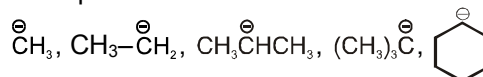
D1: A carbon intermediate which contain three bond pair and a negative charge on it, is called carbanion.

Hybridisation: Hybridisation of carbanion may be sp^3 , sp^2 or sp .

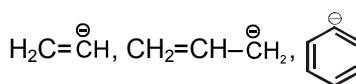
Hybridisation

Example

sp^3



sp^2



sp

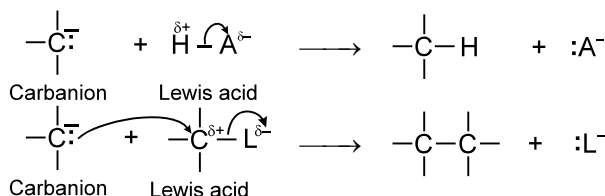
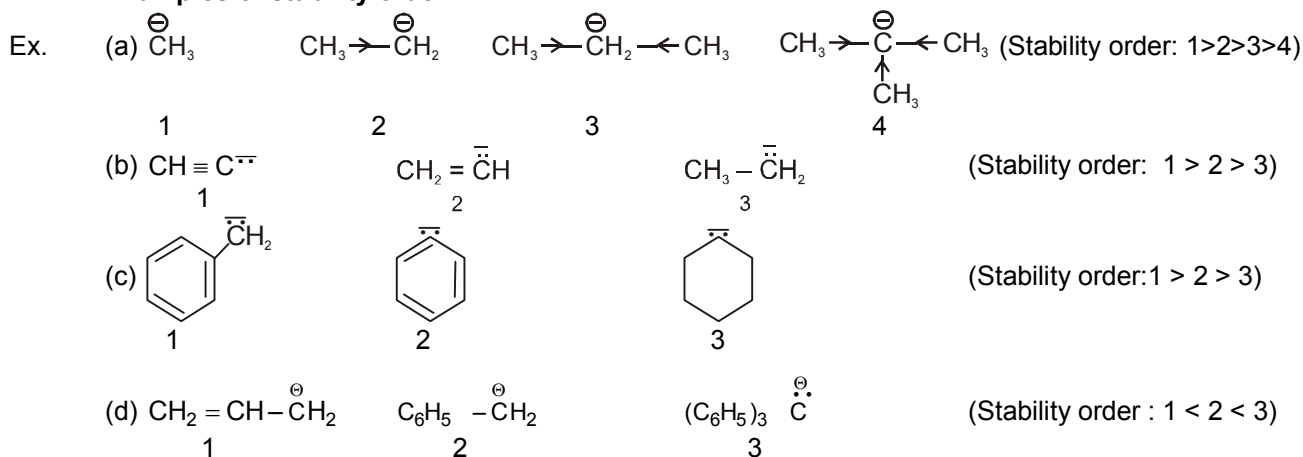


Stability of carbanion: Carbanions are stabilised by electron withdrawing effect as

- (i) – I effect (ii) –M effect (iii) Delocalisation of charge

Carbanions are Lewis bases. In their reactions they seek a proton or some other positive centre to which they can donate their electron pair and thereby neutralize their negative charge.



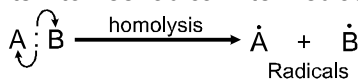
**Examples of stability order:**

Rearrangement: Generally carbanions do not undergo rearrangement.

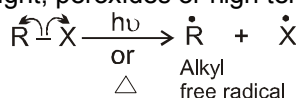
Section (B) : Carbon free radicals

D2: Free radical: An uncharged intermediate which has three bond pair and an unpaired electron on carbon.

Homolysis of covalent bond results into free radical intermediates possess the unpaired electrons.

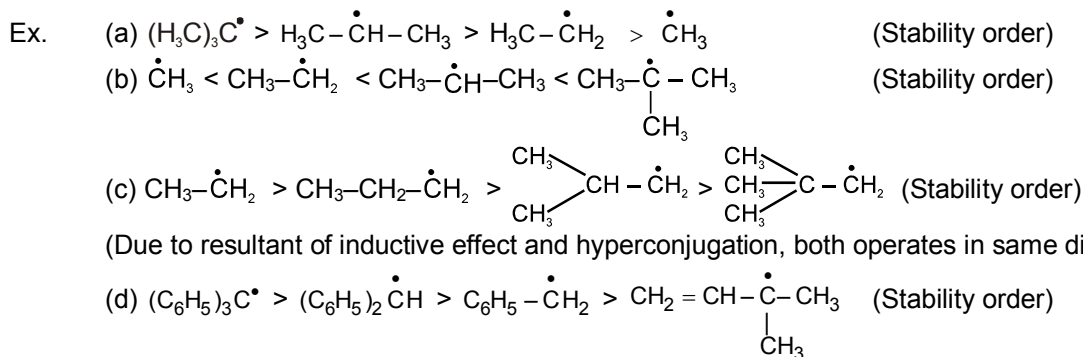


It is generated in presence of sun light, peroxides or high temperature



- Note:**
- (i) It is neutral species with odd e^- .
 - (ii) It is paramagnetic in nature due to odd e^- .
 - (iii) Rearrangement is not observed generally.
 - (iv) Carbon atom having odd electron is in sp^2 hybridised state
 - (v) Any reaction if it is carried out in the presence of sunlight, peroxide or high temperature it generally proceeds via free radical intermediate.

Stability of free radical: It is stabilised by resonance, hyperconjugation and + I groups.





Section (C) : Carbocations

D3: A carbon intermediate which contains three bond pairs & a positive charge on it is called carbocation.

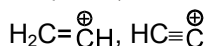
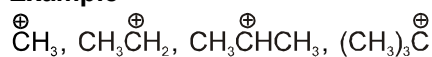
Hybridisation: Carbocation may be sp^2 or sp hybridized.

Hybridisation

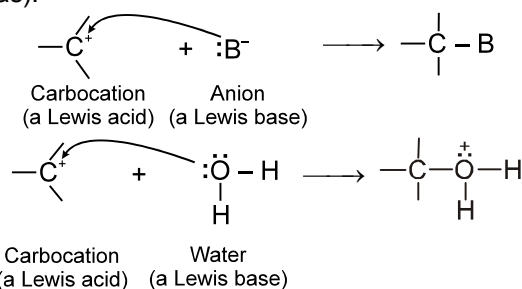
sp^2

sp

Example



Carbocations are electron deficient. They have only six electrons in their valence shell, and because of this, carbocations act as Lewis acids. Most of the carbocations are short-lived and highly reactive, they occur as intermediates in some organic reactions. Carbocations react with Lewis bases or ions that can donate the electron pair that they need to achieve a stable octet of electrons (i.e., the electronic configuration of a noble gas):



Because carbocations are electron seeking reagents, chemists call them electrophiles. All Lewis acids, including protons, are electrophiles. By accepting an electron pair, a proton achieves the valence shell configuration of helium; carbocations achieve the valence shell configuration of Neon.

Stability: Carbocations are stabilised by

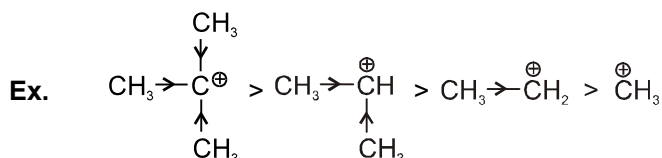
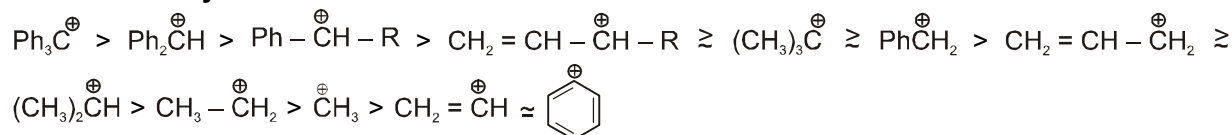
(i) +M effect

(ii) Delocalisation of charge

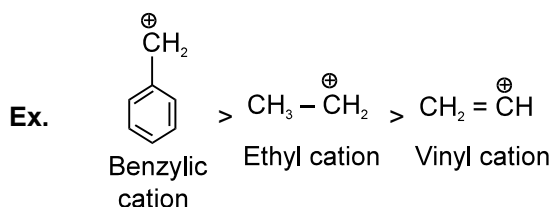
(iii) Hyperconjugation

(iv) +I effect

General stability order:



t-Butyl carbocation has +I effect of three Me-groups and also Hyperconjugation effect which makes it most stable.



In Benzylic cation, extensive resonance is seen which stabilises C^+ .

In Ethyl carbocation +I and hyperconjugation of Me-group stabilizes carbocation.

In vinyl carbocation stability decreases rapidly since carbon of (CH_2) is sp^2 hybridized which is slightly more electronegative hence acts as -I group which increases (+) charge density.



Rearrangement of carbocations:

Whenever an Intermediate carbocation is formed in reaction it may rearranges.

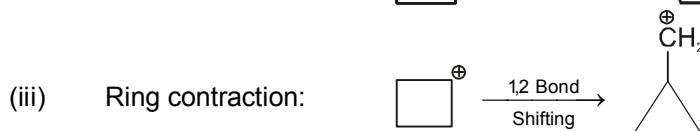
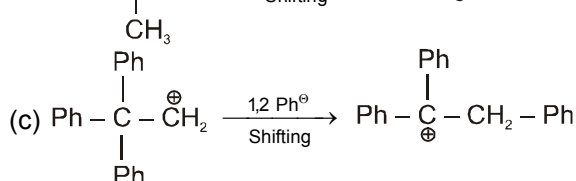
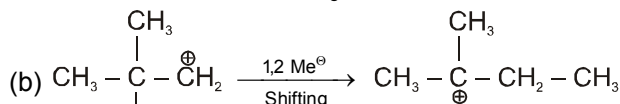
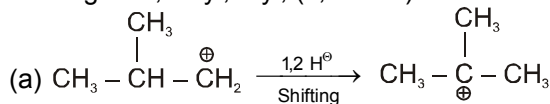
Only those carbocation will rearrange which can produce more stable species. It can be done either by

(i) Shifting of H, alkyl, aryl, bond (1, 2 shifting)

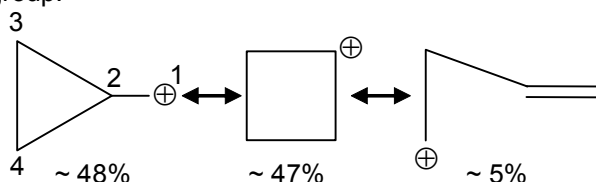
(ii) Ring expansion (more strained ring to less strained ring)

(iii) Ring contraction

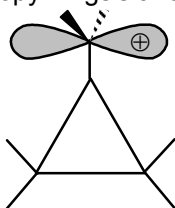
(i) Shifting of H, alkyl, aryl, (1, 2 shift)



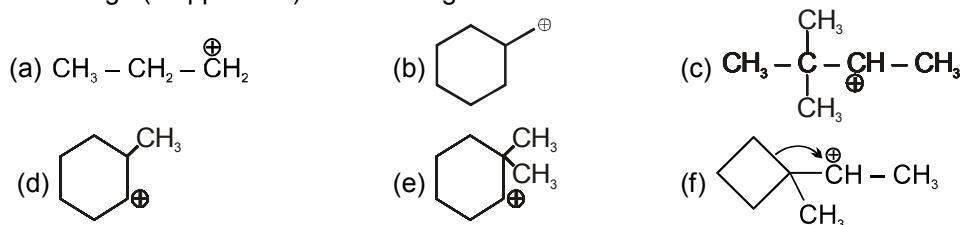
Note: Cyclopropylmethyl cation: In solvolysis of simple primary cyclopropylmethyl systems the rate is enhanced because of the participation by the σ -bonds of the rings. The ion that forms initially is an unarranged cyclopropylmethyl cation that is symmetrically stabilized, that is, both the 2, 3 and 2, 4 σ -bonds help stabilize the positive charge. Cyclopropyl group stabilizes on adjacent positive charge even better than a phenyl group.



This special stability, which increases with each additional cyclopropyl group, is a result of conjugation between the bend orbitals of the cyclopropyl rings and the vacant P-orbital of cationic carbon.



Q. Rearrange (if applicable) the following carbocations into more stable form:



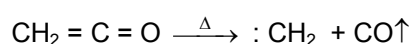
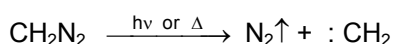
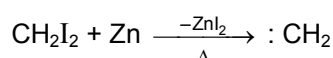
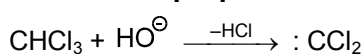


	Carbon free radical	Carbocation	Carbanion
Shape	trigonal planar	trigonal planar	Pyramidal
Hybridisation	sp^2	sp^2	sp^3
No. of electrons in outermost shell	7	6	8

D4: Carbenes (Divalent Carbon intermediates) :

There is a group of intermediates in which carbon forms only two bonds. These neutral divalent carbon species are called carbenes. Most carbenes are highly unstable that are capable of only fleeting existence. Soon after carbenes are formed, they usually react with another molecules.

Methods of preparation of carbene :

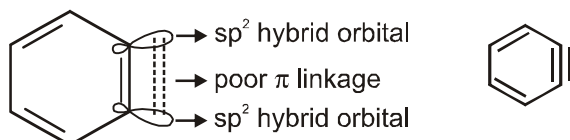


Types of carbene	Singlet	Triplet $-\dot{\text{C}}-$
Shape	Bent	Linear
Hybridisation	sp^2	sp
Nature of reaction	stereospecific	None
State	Excited state	Ground state
Magnetic	Diamagnetic	Paramagnetic
Nature	Paired electrons	Diradical

D5: Nitrenes: The nitrogen analog of carbenes are nitrenes. They are very much reactive since in them octet of N is incomplete. In nitrenes only one valencies of N are satisfied.



D6: Benzyne : The benzene ring has one extra C–C π bond in benzyne



Clearly, we can see that the newly formed π bond cannot enter in resonance with other π orbitals of ring. Since it is in perpendicular plane.

It is also important to note that hybridisation of each carbon involved in 'Benzynic bond' is sp^2 since the overlap between these sp^2 hybrid orbitals is not so much effective.



Section (D) : Basic strength

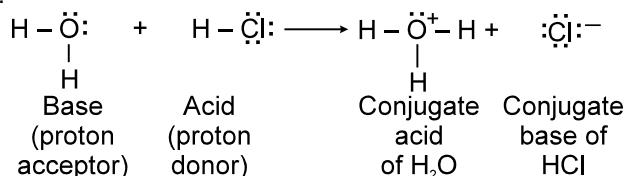
Th2. Bases

D7: (a) Arrhenius base: Base is a substance that can donate (or loose) OH^- ions in H_2O .

D8: (b) The Bronsted Lowry definition of acids and bases:

An acid is a substance that can donate (or loose) a proton, and a base is a substance that can accept a proton.

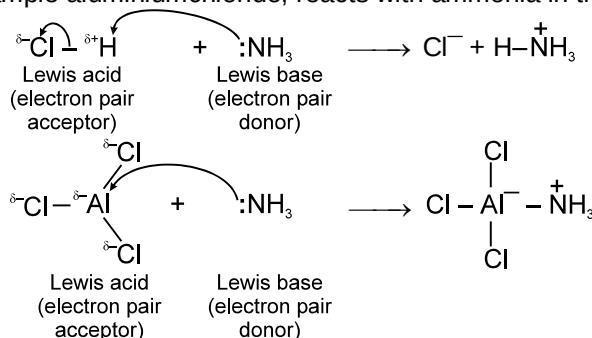
Let us consider, an example of this concept, the reaction that occurs when gaseous hydrogen chloride dissolves in water :



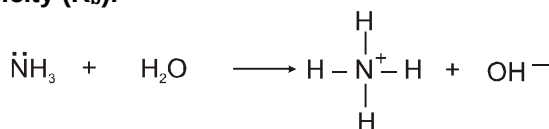
D9: (c) The Lewis definition of acids and bases:

Lewis proposed that acids are electron pair acceptors and bases are electron pair donors.

For example aluminiumchloride, reacts with ammonia in the same way that a proton donor does.



(d) Basicity (K_b):



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

$$\text{p}K_b = -\log K_b$$

Basicity order in periodic table:

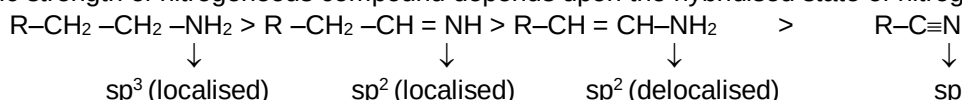
- (1) Basic strength decreases down the group,
- (2) Basic strength decreases along the period because electron neagativity increases, so electron donor tendency decreases.

1.1 Aliphatic bases:

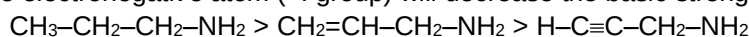
(1) On the basis of +I effect basic strength of amines should be $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$

But this order is applicable only when the amines are in gaseous state or in case of non-polar aprotic solvent.

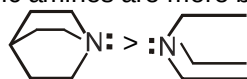
(2) Basic strength of nitrogeneous compound depends upon the hybridised state of nitrogen



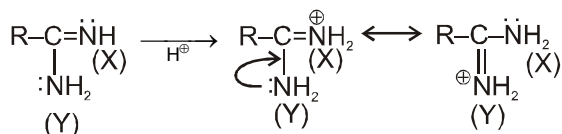
(3) More electronegative atom (–I group) will decrease the basic strength



(4) Cyclic amines are more basic than acyclic amines of same nature



(5) Amidines are more basic in nature because their conjugate acid are more stable due to resonance.

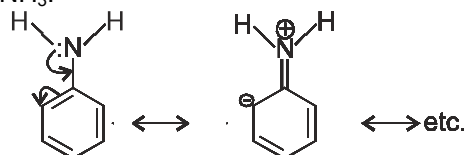


Nitrogen (X) is more basic than nitrogen (Y).

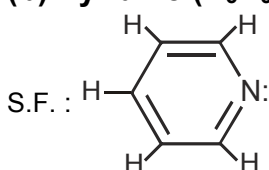
Th3. Basic strength of aromatic amines and substituted anilines:

(a) Aniline:

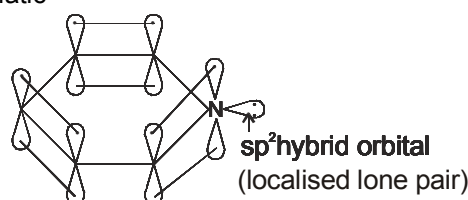
Lone pair of aniline lies in conjugation with a multiple bond, it resides in '2p' atomic orbital, so that it can get resonance stabilisation and hence, basic strength decreases. So, Aniline is a weaker base than NH_3 .



(b) Pyridine ($\text{C}_5\text{H}_5\text{N}$) : 6- π electrons, aromatic

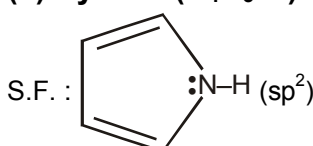


A.O. Diagram:

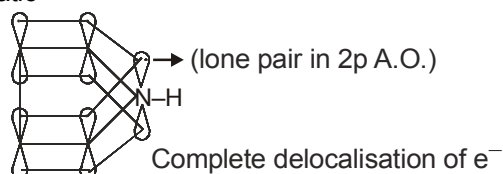


Lone pair of N in pyridine is localised so it is more basic than aniline.

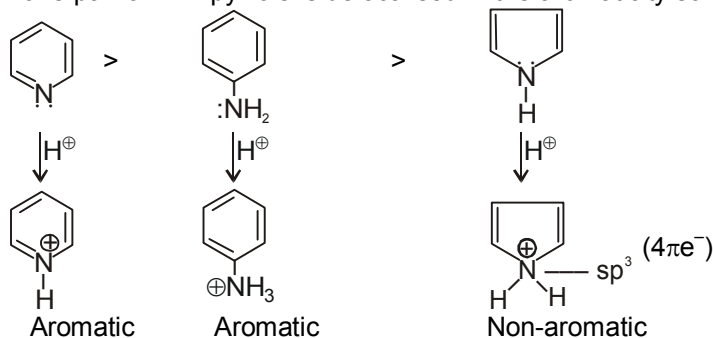
(c) Pyrrole ($\text{C}_4\text{H}_5\text{N}$) : 6- π electrons, aromatic



A.O. Diagram:



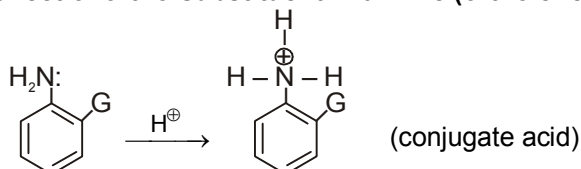
Lone pair of N in pyrrole is delocalised in the aromaticity so it is very less basic than aniline.



(d) Ortho substituted anilines :

Electron releasing groups (ERG) +M, HC, +I increases the K_b and
Electron withdrawing groups (EWG) -m, -I decreases the K_b

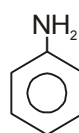
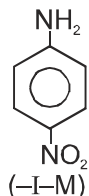
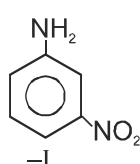
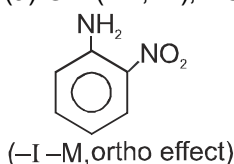
Steric effect of ortho-substituent in aniline (ortho effect) :





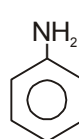
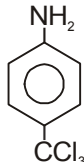
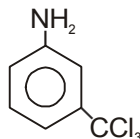
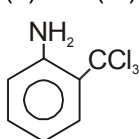
- (i) Ortho-substituted anilines are mostly weaker bases than aniline itself.
 (ii) Ortho-substituent causes steric hinderance to solvation in the product (conjugate acid i.e. cation).
 (iii) The small groups like $-NH_2$ or $-OH$ do not experience (SIR) due to small size.

Ex. (a) $G = (-M, -I); NO_2$



(Aniline > m > p > o).

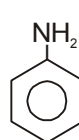
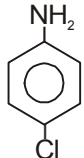
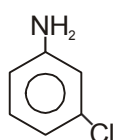
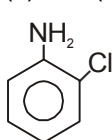
(b) $G = (-I); CCl_3$



(Aniline > p > m > o).

Only (-I) decides the order.

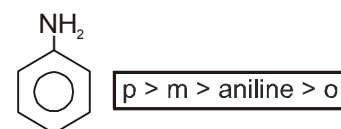
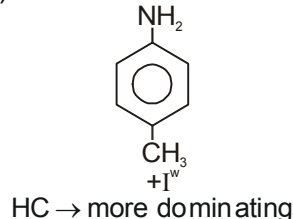
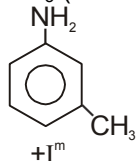
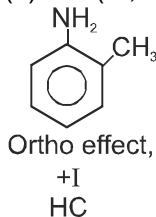
(c) $G = (-I > +m); Cl$



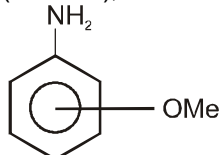
(Aniline > p > m > o)

Only (-I) decides the order.

(d) $G = (+I, HC);$ If $R = -CH_3$ (Toluidines)

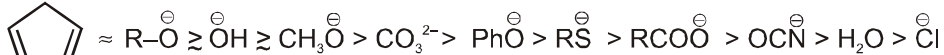
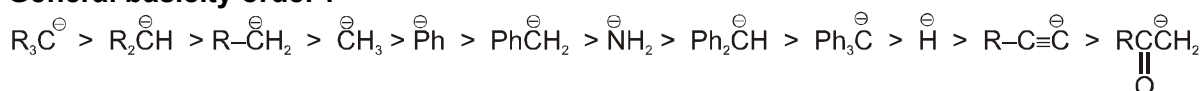


(e) $G = (+M > -I);$



K_b order : p > Aniline > o > m

General basicity order :



Th4. Solvent effect in bases :

The trend is not regular in the aqueous state as evident by their pK_b values given in Table.

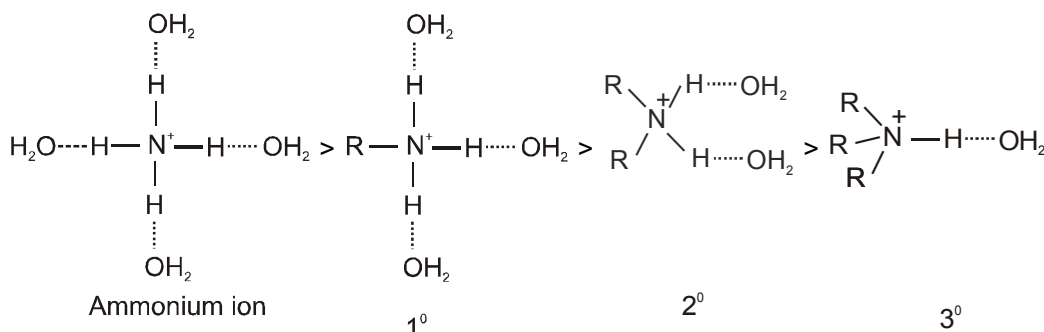
Name of amine	pK_b	Name of amine	pK_b
Methanamine	3.38	N,N-Diethylethanamine	3.25
N-Methylethanamine	3.27	Phenylmethanamine	4.70
N, N-Dimethylethanamine	4.22	Aniline	9.38
Ethanamine	3.29	N-Methylaniline	9.30
N-Ethylethanamine	3.00	N,N-Dimethylaniline	8.92

Table : pK_b Values of Amines in Aqueous Phase (Ref. NCERT)





In the aqueous phase, the substituted ammonium cations get stabilised not only by electron releasing effect of the alkyl group (+I) but also by solvation with water molecules. The greater the size of the ion (Alkyl groups are hydrophobic and inhibits H bonding and solvation.), lesser will be the solvation and the less stabilised is the ion. The order of solvation of ions are as follows:



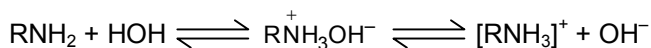
Greater is the stability of the substituted ammonium cation, stronger should be the corresponding amine as a base.

On the basis of above two sequences, we can say that the basic strength of amines is the combined effect of inductive effect, steric hindrance and solvation.

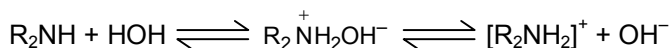


Th5. Reactions of bases :

- (i) **Nature of aqueous solution:** Amines combine with water to form alkyl ammonium hydroxides. This gives hydroxide ions in solution, thus the aqueous solution of amines is basic in nature.

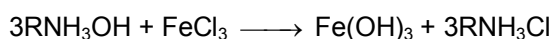


1° Amine



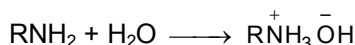
2° Amine

The aqueous solution of amines behave like NH_4OH and gives the precipitate of ferric hydroxide with ferric chloride.

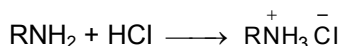


Brown ppt.

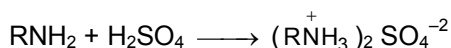
- (ii) **Aliphatic and aromatic amines form salt because of their basic nature:**



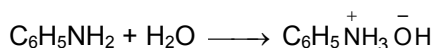
Alkylammoniumhydroxide



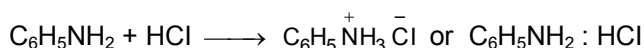
Alkylammoniumchloride



Similarly we get



Aniliniumhydroxide



Aniliniumchloride Anilinehydrochloride

Salts of amines are ionic compounds and hence water soluble.





Section (E) : Acidic strength

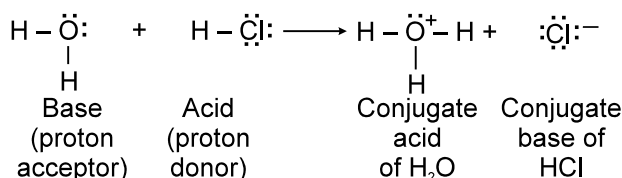
Th6. Acids

D10: (a) Arrhenius acid: An acid is a substance that can donate (or loose) a proton in H_2O .

D11: (b) The Bronsted Lowry definition of acids and bases:

An acid is a substance that can donate (or loose) a proton, and a base is a substance that can accept a proton.

Let us consider, an example of this concept, the reaction that occurs when gaseous hydrogen chloride dissolves in water :

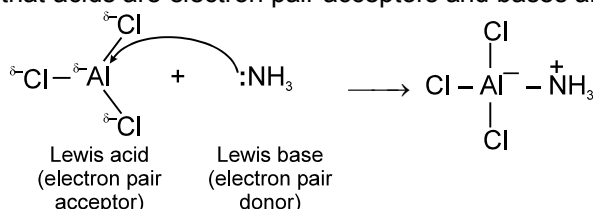


Hydrogen chloride, a very strong acid, transfers its proton to water. Water acts as a base and accepts the proton. The products that result from this reaction are a hydronium ion (H_3O^+) and a chloride ion (Cl^-).

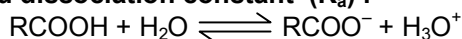
The molecule or ion that forms when an acid loses its proton is called the conjugate base of that acid. (The chloride ion is the conjugate base of HCl). The molecule or ion that is formed when a base accepts a proton is called the conjugate acid of that base.

D12: (c) The Lewis definition of acids and bases

Lewis proposed that acids are electron pair acceptors and bases are electron pair donors.



(d) Acid dissociation constant (K_a) :



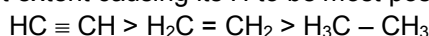
$$K_{eq} = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}][\text{H}_2\text{O}]}$$

$$K_a = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}]}$$

$$\text{p}K_a = -\log K_a$$

6.1 Relative acidity of hydrocarbons :

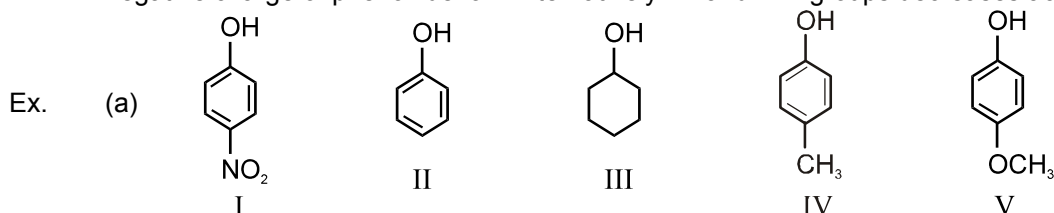
Being most electronegative the sp hybridised carbon atom of ethyne polarizes its C-H bond to the greatest extent causing its H to be most positive therefore ethyne is most acidic hydrocarbon.



6.2 Acidity of phenols :

The phenoxide ion is more stabilised by resonance than the unionised phenol.

Groups which are $-I$, $-m$ increases acidic character of phenol because effectively dispersing the negative charge of phenoxide ion. Alternatively $+I$ and $+m$ groups decreases acid strength.



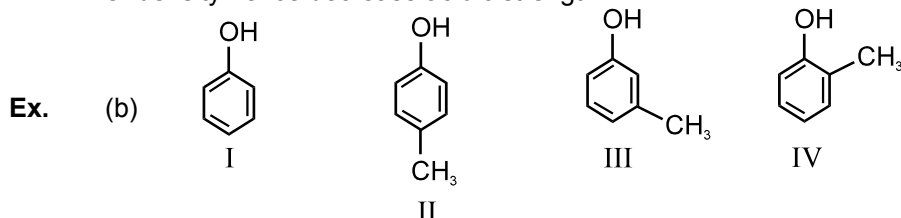
Ans. Acid strength order : I > II > IV > V > III



Sol. Step-1. III will be least acidic as it has no dispersion of negative charge (No delocalisation of negative charge).

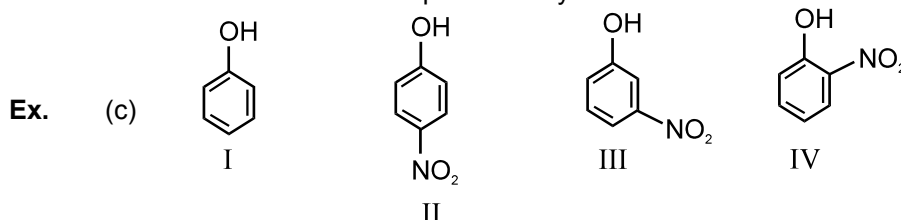
Step-2. Since $-I$, $-M$ group will increase acid strength, Nitrophenol will be most acidic followed by phenol,

Step-3. Amongst cresol and methoxyphenol, methoxyphenol has $+M$ effect of $-OCH_3$ which increases e^- density hence decrease acidic strength



Ans. Acid strength order: $I > III > II > IV$

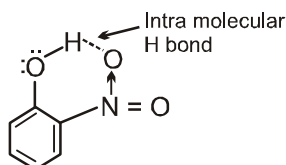
Sol. Step-1: Notice that CH_3 have $+I$ effect so all methylphenols (cresols) are less acidic than phenol (I).
Step-2: Now amongst cresols p- and o- CH_3 are increasing the e^- density due to their hyper conjugation but ortho isomer has viable $+I$ effect also, which will help in destabilising phenoxide ion therefore o- is least acidic. Since at meta position only $+I$ works it as least e^- density amongst the cresol.



Ans. Acid strength order: $II > IV > III > I$

Sol. Step-1 : In nitrophenols $-I$ effect of NO_2 will help to increase acidic strength hence phenol is least acidic amongst all nitrophenols

Step-2 : Only $-I$ effect is applicable in meta nitrophenol it will be number three. Now $-o$, $-p$ have both $-I$ and $-M$ effect of NO_2 group over OH and in this particular case para isomer is more acidic than ortho since



H is trapped by NO_2 group.

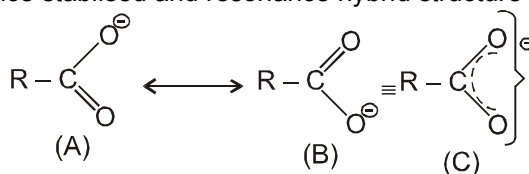
Table : pK_a values of some phenols and Ethanol. (Ref. NCERT)

Compound	Formula	pK_a	Compound	Formula	pK_a
o-Nitrophenol	$o-O_2N-C_6H_4-OH$	7.2	o-Cresol	$o-CH_3-C_6H_4-OH$	10.2
m-Nitrophenol	$m-O_2N-C_6H_4-OH$	8.3	m-Cresol	$m-CH_3-C_6H_4-OH$	10.1
p-Nitrophenol	$p-O_2N-C_6H_4-OH$	7.1	p-Cresol	$p-CH_3-C_6H_4-OH$	10.2
Phenol	C_6H_5-OH	10	Ethanol	C_2H_5OH	15.9

From the above data, you will note that phenol is million times more acidic than ethanol.

6.3 Acidity of carboxylic acids :

Conjugate base of carboxylic acid exists as two equivalent canonical structures (A) and (B). This ion is resonance stabilised and resonance hybrid structure is (C).

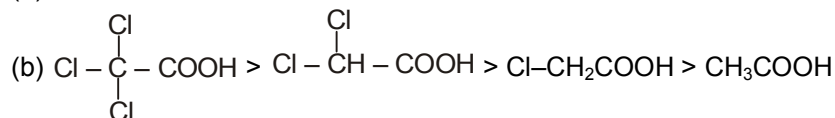


Electron withdrawing group ($-M$, $-I$ effect) increases acidic nature.

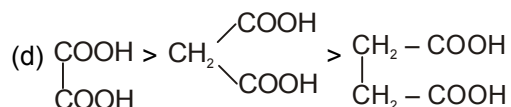
Electron releasing group ($+M$, $+I$ effect) decreases acidic nature.



Ex. (a) $\text{F}-\text{CH}_2-\text{COOH} > \text{Cl}-\text{CH}_2\text{COOH} > \text{Br}-\text{CH}_2\text{COOH} > \text{I}-\text{CH}_2\text{COOH}$

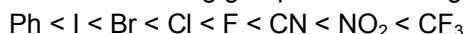


(c) $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{CH}_3-\text{CH}_2-\text{COOH}$

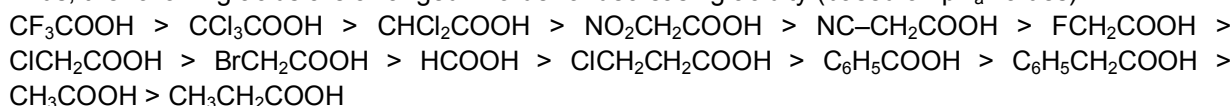


The effect of the following groups in increasing acidity order is

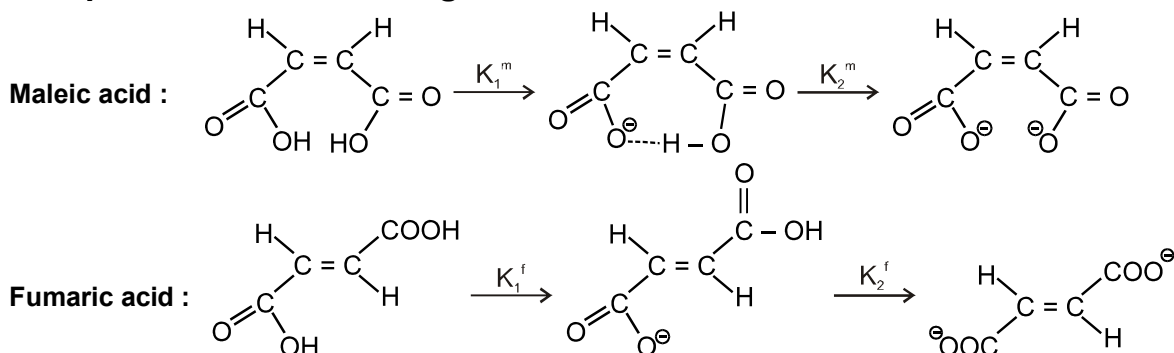
[Ref. NCERT]



Thus, the following acids are arranged in order of decreasing acidity (based on pK_a values) :



6.4 Comparison between two geometrical isomers :



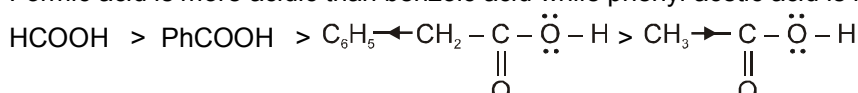
Now $K_1^m > K_1^f$

Since the conjugate base is stabilised by intramolecular H bonding.

But $K_2^f > K_2^m$ Since in maleate ion, after donation of H^+ two $-\text{COO}^-$ groups faces each other and makes system unstable. In fumarate ion this repulsion is minimised.

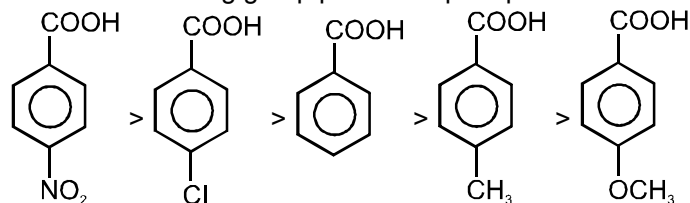
6.5 Acidic strength of substituted benzoic acids :

Formic acid is more acidic than benzoic acid while phenyl acetic acid is more acidic than acetic acid.

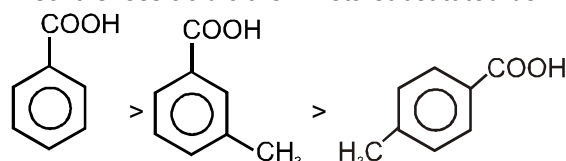


Electron withdrawing group attached to benzene ring will increase the acidic strength while electron releasing group decreases acidic strength.

If electron donating group present at para position than it is always less acidic than benzoic acid.

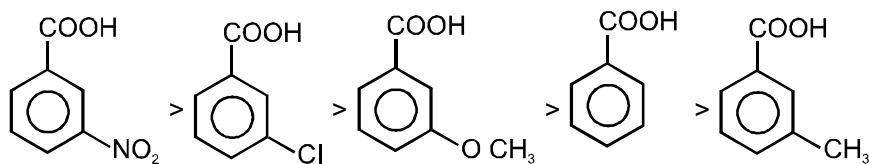


Also it is less acidic than meta substituted benzoic acid.



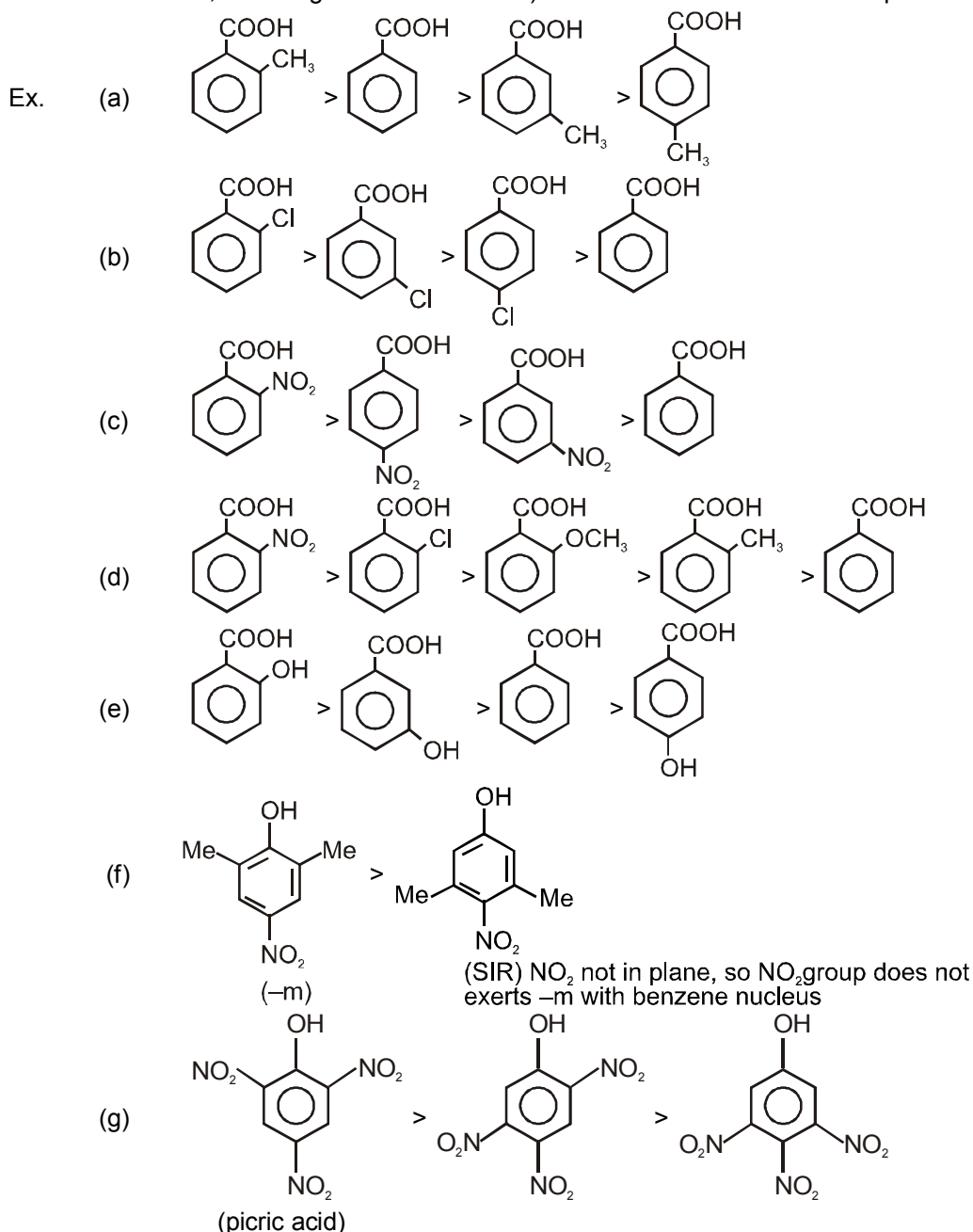


On the other hand if e^- withdrawing group is present at meta position then it is more acidic than benzoic acid.



Th7. Ortho effect :

D13. It is common observation that generally ortho substituted benzoic acids are more acidic as compared to their isomers and benzoic acids itself. This is called ortho effect (which is combined effect of steric hindrance, crowding & electronic effect) in benzoic acid. However exceptions are seen.





Section (F) : Feasible reactions of acids and bases

Th8. Reaction of acid with salt:



Remark: A stronger acid displaces the weaker acid from weak acid metal salt. The weaker acid is released out as a gas or liquid or precipitates out as a solid. The weaker acid cannot displace the stronger acid from the salt.

1. $2 \text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$
2. $\text{Na}_2\text{SO}_4 + 2\text{HCl} \longrightarrow \text{No reaction}$
3. $\text{CH}_3\text{COONa} + \text{CH}_3\text{SO}_3\text{H} \longrightarrow \text{CH}_3\text{COOH} + \text{CH}_3\text{SO}_3\text{Na}$ (feasible)
4. $\text{CH}_3\text{COONa} + \text{PhOH} \longrightarrow \text{PhONa} + \text{CH}_3\text{COOH}$ (not feasible)

Section (G) : Tautomerism

Th9. Tautomerism

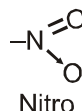
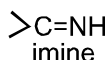
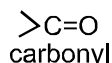
D14: Definition :

Tautomerism is a phenomenon by which a single compound exists in two or more readily interconvertible structures that differ in the relative positions of at least one atomic nucleus, generally hydrogen.

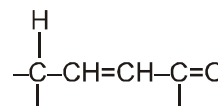
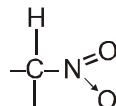
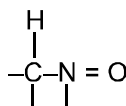
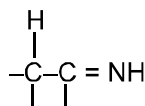
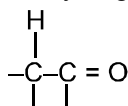
These two isomers remain in dynamic equilibrium and can be isolated also give different lab test.

Conditions :

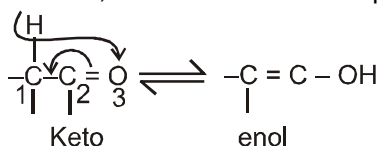
1. Usually present in the following functional groups



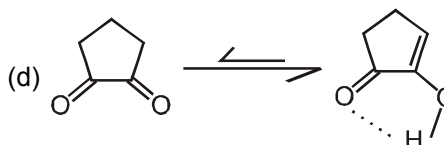
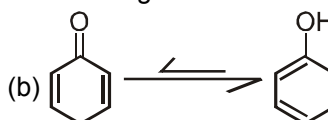
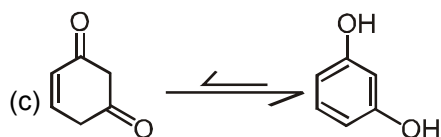
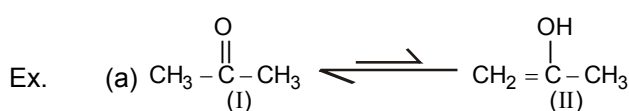
2. Basic need for its existence is attachment of these groups with the sp^3 hybridised C-atom having atleast one hydrogen atom as –



To get tautomer of above structures α -hydrogen atom is shifting to more electronegative atom attached to double bond (i.e. hydrogen atom from 1st atom to 3rd atom) and double bond is developed between 1,2-atom from 2,3-atom. This can be represented as:



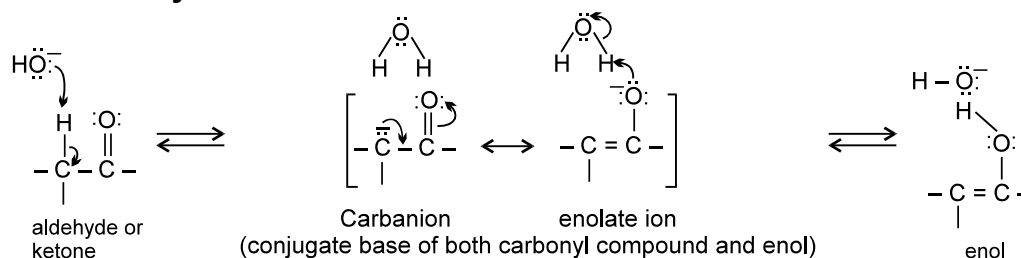
These two forms (remain in equilibrium) are called tautomers of each other. The interconvertibility of tautomers is a chemical reaction which involves making and breaking of bonds.





Th10. Keto-enol tautomerisation :

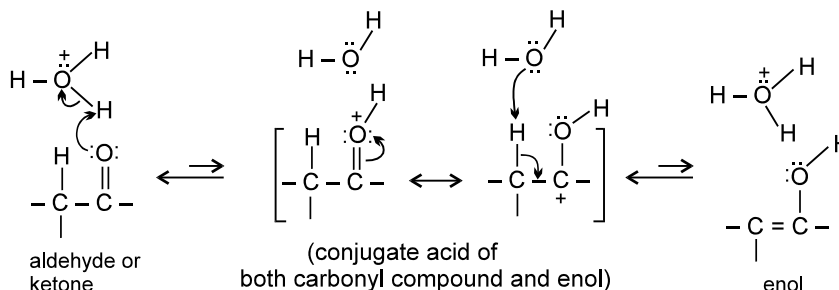
10.1 (A) Base-catalyzed enolization :



Protonation of the carbanion by water on the α -carbon gives back the carbonyl compound. Protonation on oxygen gives the enol. Notice that the enolate ion is the conjugate base of both the carbonyl compound and the enol.

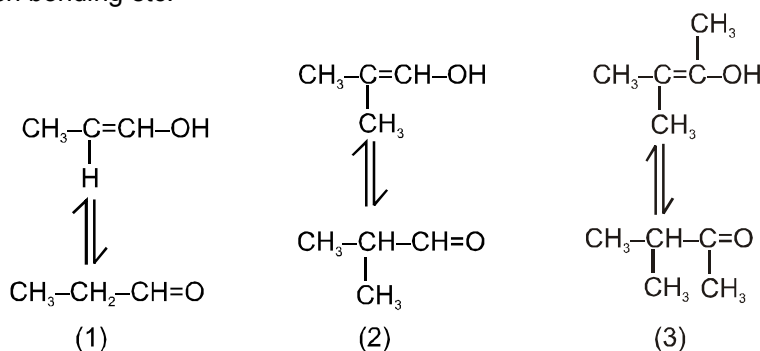
(B) Acid-catalyzed enolization :

Involves the conjugate acid of the carbonyl compound. Recall that this ion has carbocation characteristics. Loss of the proton from oxygen gives back the starting carbonyl compound; loss of the proton from the α -carbon gives the enol. Notice that an enol and its carbonyl isomer have the same conjugate acid.



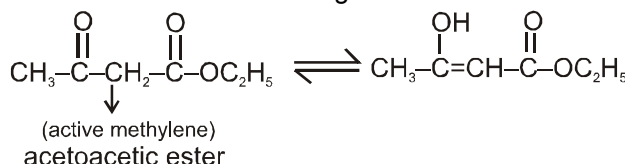
10.2 % Enol content in the carbonyl compounds :

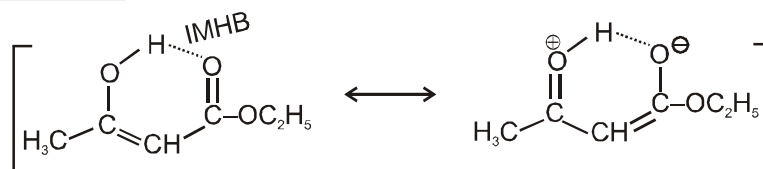
- (a) For monocarbonyl % Enol is very less
 (b) Enol content increases with increase in the stability of enol by resonance, hyperconjugation, hydrogen bonding etc.



Decreasing order of enol content for above carbonyl compounds is : 3 > 2 > 1.

- (c) For a carbonyl compound having active methylene group percentage of enol content will be more because enolic form has intramolecular H-bonding and also it will be stabilised by resonance.





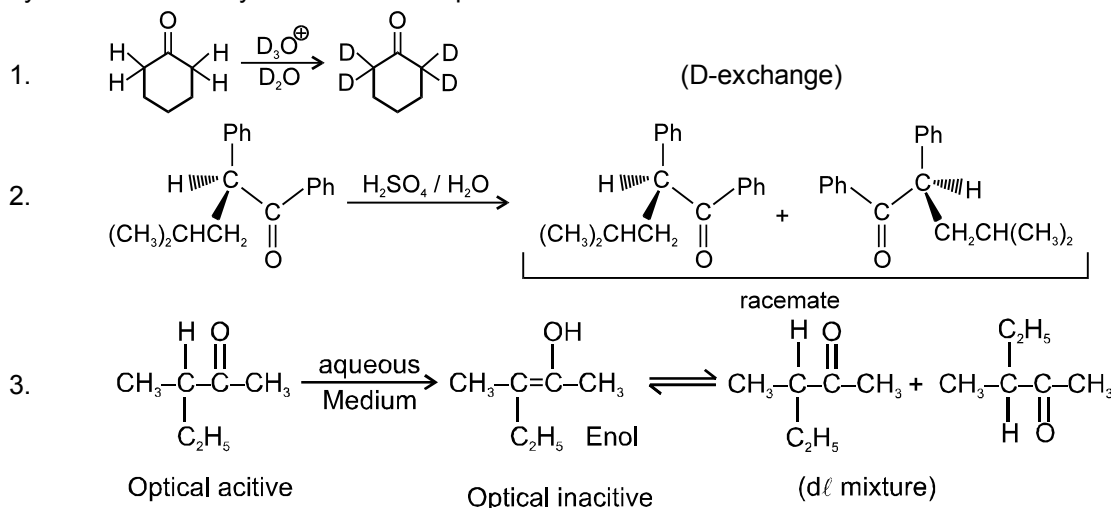
(d) If active methylene group is more acidic then enol content will be more.

For example in acetyl acetone ($\text{CH}_3\text{C}(=\text{O})\text{CH}_2\text{C}(=\text{O})\text{CH}_3$) enolic content is 75–76% while it is 7–8% in acetoacetic ester because ester group shows less electron withdrawing nature than keto group.

(e) Percentage of enol content is more in non-polar media while % of keto form is more in polar media.

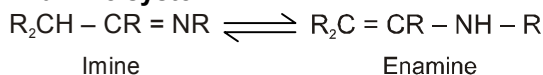
Th11. Racemisation and D-exchange :

Exchange of α -hydrogen from deuterium as well as racemization at the α -carbon are catalyzed not only by bases but also by acids due to the phenomenon of tautomerisation.



Th12. Other examples of tautomers (*not to be done in class only for the reference of students*)

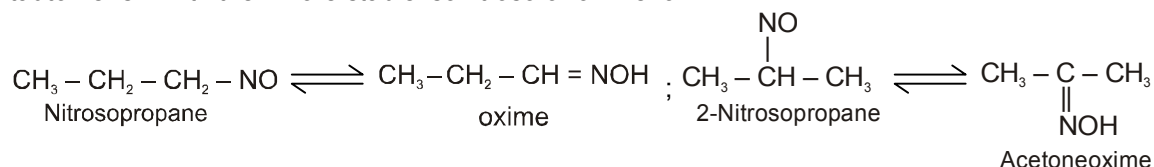
(a) Imine-Enamine system :



Among these two tautomers, enamines are stable only when there is no hydrogen on the nitrogen, otherwise the imine form predominates.

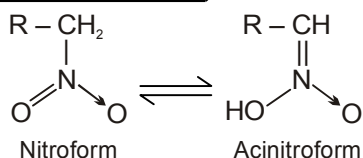
(b) Nitroso-oxime system:

Like primary and secondary nitro compounds, primary and secondary nitroso compounds also exhibit tautomerism with their more stable isonitroso or oxime form.

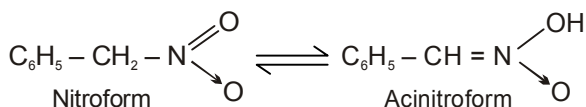


(c) Nitro-Acinitro system:

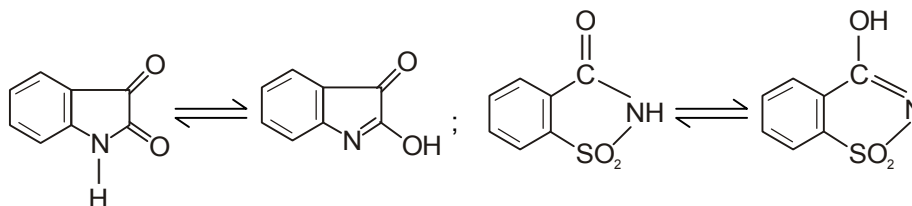
The acidic nature of the nitro compounds gives rise to the belief that the nitro compounds exist in two forms, a more stable or normal nitro form and the less stable acinitro form.



The stability of the nitro form is more as compared to the acinitro form because it is stabilised by resonance.



(d) Lactam-Lactim system :



CHECK LIST

Definitions (D)			Theories (Th)		
D1 :	Carbanion	☐	Th1 :	Reaction intermediates	☐
D2 :	Free radicals	☐	Th2 :	Bases	☐
D3 :	Carbocation	☐	Th3 :	Basic strength of aromatic amines and substituted anilines	☐
D4 :	Carbenes (Divalent carbon intermediates)	☐	Th4 :	Solvent effect in bases	☐
D5 :	Nitrenes	☐	Th5 :	Reactions of bases	☐
D6 :	Benzyne	☐	Th6 :	Acids	☐
D7 :	Arrhenius base	☐	Th7 :	Ortho effect	☐
D8 :	Bronsted base	☐	Th8 :	Reactions of acids with salts	☐
D9 :	Lewis definition of acids and bases	☐	Th9 :	Tautomerism	☐
D10 :	Arrhenius acid	☐	Th10 :	Keto-enol tautomerisation	☐
D11 :	Bronsted acid	☐	Th11 :	Racemisation and D-exchange	☐
D12 :	Lewis definition of acids and bases	☐	Th12 :	Other examples of tautomers	☐
D13 :	Ortho effect	☐			
D14 :	Tautomerism	☐			





Exercise-1

Marked Questions may have for Revision Questions.

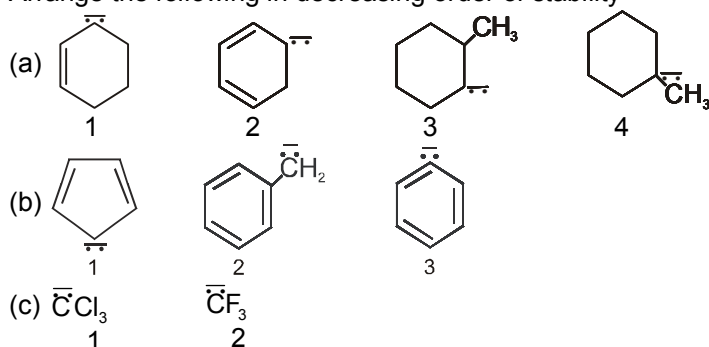
PART - I : SUBJECTIVE QUESTIONS

Section (A) : Carbanions

A-1. Arrange the following in decreasing order of stability.

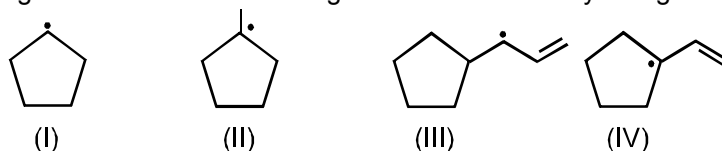
(a)	$\text{CH}_2\text{--NO}_2^-$ 1	$\text{CH}_2\text{--CHO}$ 2	$\text{CH}\equiv\text{C}^-$ 3
(b)	$\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--}\overset{\ominus}{\text{C}}\text{H}_2$ 1	$\text{CH}_3\text{--}\overset{\ominus}{\text{C}}\text{H--CH}_2\text{CH}_3$ 2	$(\text{CH}_3)_2\overset{\ominus}{\text{C}}\text{--CH}_2\text{CH}_3$ 3

A-2. Arrange the following in decreasing order of stability

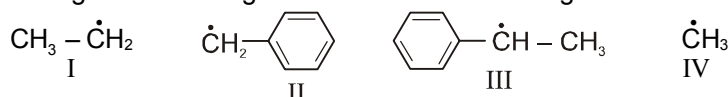


Section (B) : Carbon free radicals

B-1. Rank the following free radicals in increasing order of their stability and give appropriate reasons.

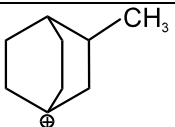
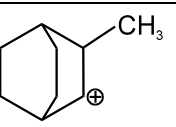
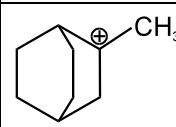
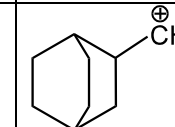


B-2. Arrange the following free radicals in decreasing order of stability :



Section (C) : Carbocations

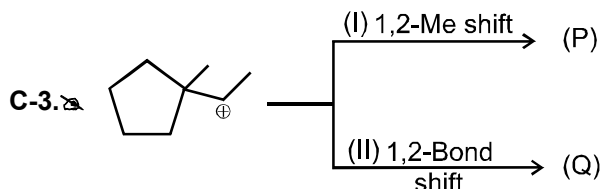
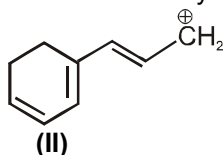
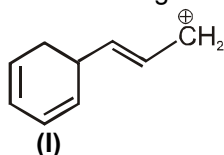
C-1. Arrange the following carbocations in decreasing order of their stability :

	(I)	(II)	(III)	(IV)
(P)	$\text{CH}_3\text{--CH}_2\text{--}\overset{\oplus}{\text{C}}\text{H}_2$	$\text{CH}_3\text{--CH--}\overset{\oplus}{\text{C}}\text{H}_2$ F	$\text{CH}_3\text{--CH--}\overset{\oplus}{\text{C}}\text{H}_2$ Br	$\text{CH}_3\text{--CH--}\overset{\oplus}{\text{C}}\text{H}_2$ Cl
(Q)	$\text{CH}_3\text{--}\overset{\oplus}{\text{C}}\text{H--C}_2\text{H}_5$	$\text{CH}_3\text{--}\overset{\oplus}{\text{C}}\text{(CH}_3\text{)}_2$	$\text{Ph--}\overset{\oplus}{\text{C}}\text{(CH}_3\text{)--C}_2\text{H}_5$	$\text{Ph--}\overset{\oplus}{\text{C}}\text{(Ph)--C}_2\text{H}_5$
(R)				





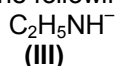
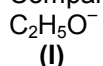
C-2. Which of the following carbocation is more stable and why ?



Draw the structures of P and Q.

Section (D) : Basic strength

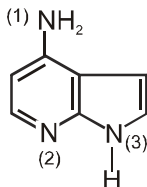
D-1. Compare the basic strength of the following compounds:



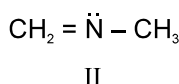
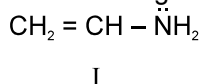
D-2. Compare the basic strength of the following compounds :

(a)	PhNH_2	Ph_2NH	Ph_3N
(b)			
(c)	$\text{CH}_3-\underset{\text{Ph}}{\text{CH}}-\text{NH}_2$	$\text{CH}_3-\text{CH}_2-\underset{\text{Ph}}{\text{N}}\text{H}$	$\text{Ph}-\text{CH}_2-\text{CH}_2-\text{NH}_2$

D-3. Which of the following group is most basic in the given compounds :

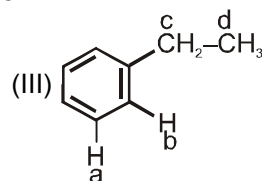
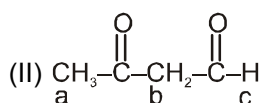
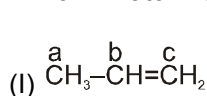


D-4. Which of the following is a stronger base ? Give reason to justify your answer.

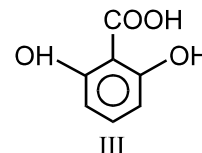
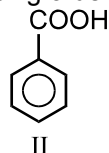
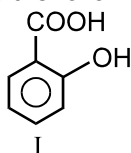


Section (E) : Acidic strength

E-1. Which 'H' atom is most acidic in the following compounds.



E-2. Arrange the following in decreasing order of acidity



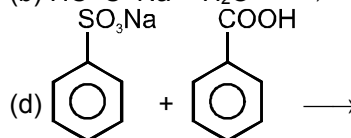
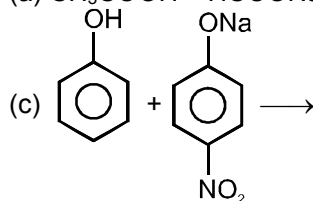
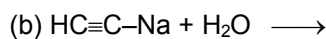
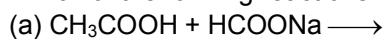
E-3. The given compound X = is a strong acid. Justify this statement.



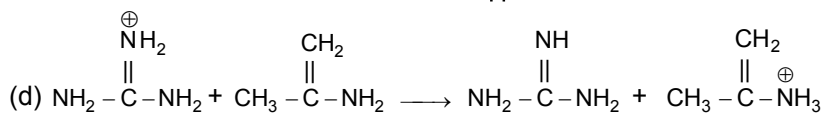
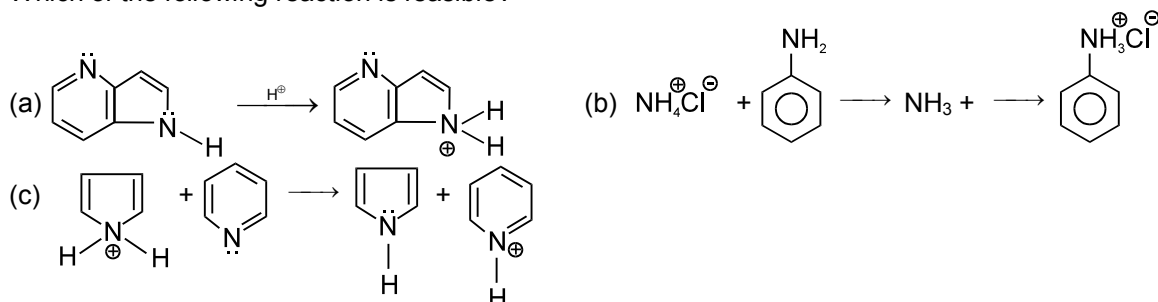


Section (F) : Feasible reactions of acids and bases

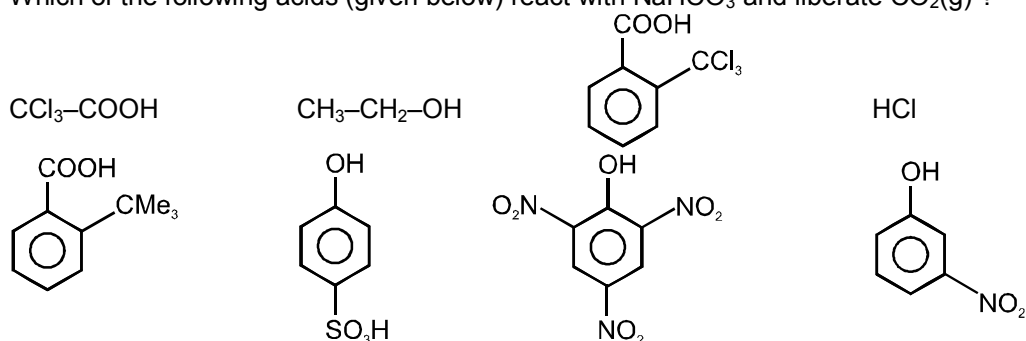
F-1. Which of the following reactions is/are feasible ?



F-2. Which of the following reaction is feasible?

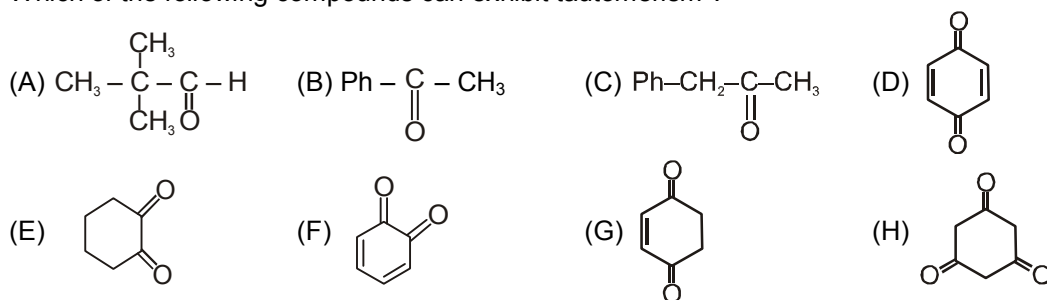


F-3. Which of the following acids (given below) react with NaHCO_3 and liberate $\text{CO}_2(\text{g})$?

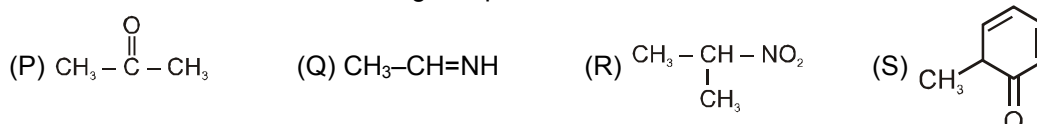


Section (G) : Tautomerism

G-1. Which of the following compounds can exhibit tautomerism ?



G-2. Write the tautomers of the following compounds :



G-3. Monocarbonyl compounds have very small percentage enol form at equilibrium. Explain.



PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : Carbanions

A-1. Which of the following is the least stable carbanion?

- (A) $\text{HC} \equiv \text{C}^-$ (B) $(\text{C}_6\text{H}_5)_3\text{C}^-$ (C) $(\text{CH}_3)_3\text{C}^-$ (D) CH_3^-

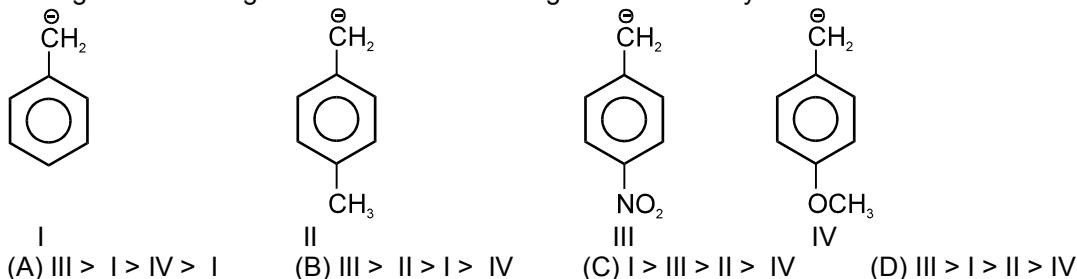
A-2. The most stable anion is :



A-3. In which of the following pairs of carbanion the first one is more stable than second.

- (A) CF_3^- , CCl_3^- (B) $\text{HC} \equiv \text{C}^-$, $\text{H}_2\text{C} = \text{C}^-$
 (C) ,  (D) $(\text{CH}_3)_3\text{C}^-$, $\text{H}_3\text{C}-\text{CH}_2^-$

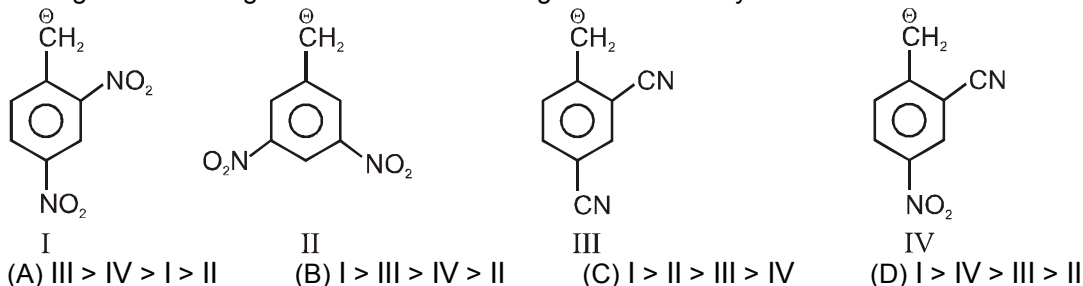
A-4. Arrange the following carbanions in decreasing order of stability :



A-5. The most stable anion is :



A-6. Arrange the following carbanions in increasing order of stability :



Section (B) : Carbon free radicals

B-1. Among the following, the paramagnetic species is :

- (A) Free radical (B) Carbocation (C) Carbanion (D) All the three

B-2. The stability of given free radicals in decreasing order is :

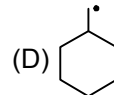
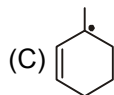
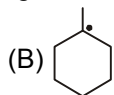
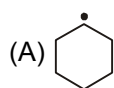
- (i) $\text{CH}_3-\dot{\text{C}}\text{H}_2$ (ii) $\text{CH}_3-\dot{\text{C}}\text{H}-\text{CH}_3$ (iii) $\text{CH}_3-\dot{\text{C}}(\text{CH}_3)-\text{CH}_3$ (iv) $\dot{\text{C}}\text{H}_3$
 (A) iii > iv > i > ii (B) i > ii > iii > iv (C) iii > ii > iv > i (D) iii > ii > i > iv

B-3. Which of the following is the correct order of stability of free radicals ?

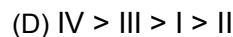
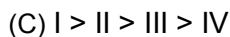
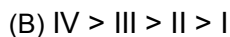
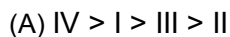
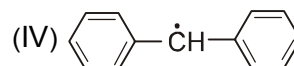
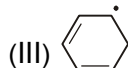
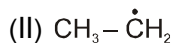
- (A) benzyl > allyl > 2° > 1° (B) allyl > benzyl > 2° > 1°
 (C) allyl > 2° > 1° > benzyl (D) benzyl > 2° > 1° > allyl



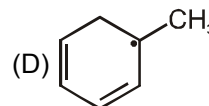
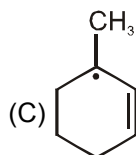
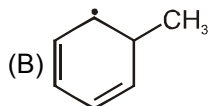
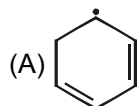
B-4. Most stable radical among the following is :



B-5. Arrange the following radicals in decreasing order of their stability.

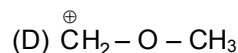
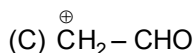
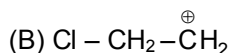
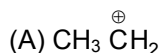


B-6. Least stable radical among the following is :

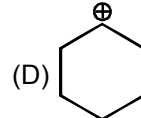
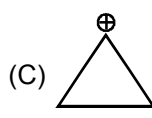
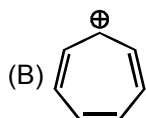
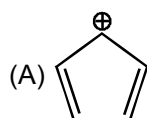


Section (C) : Carbocations

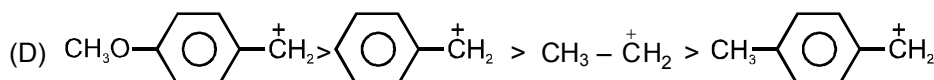
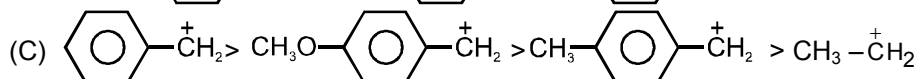
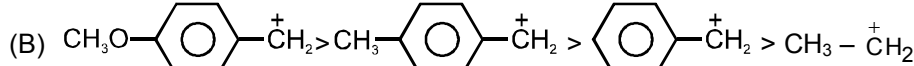
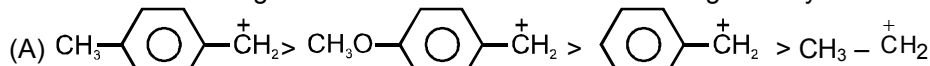
C-1. The most unstable carbocation is :



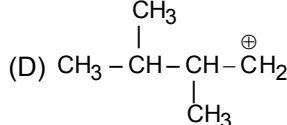
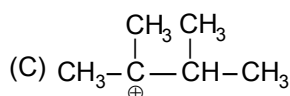
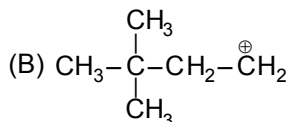
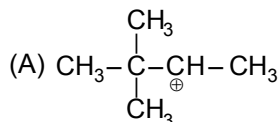
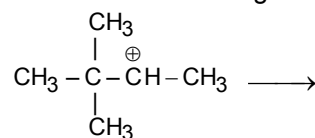
C-2. The most stable carbocation is :



C-3. Which of the following shows the correct order of decreasing stability ?

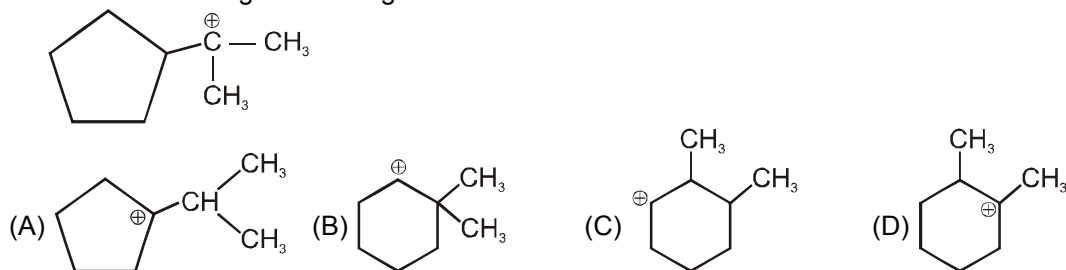


C-4. Which of the following is the rearranged more stable carbocation of the given species ?

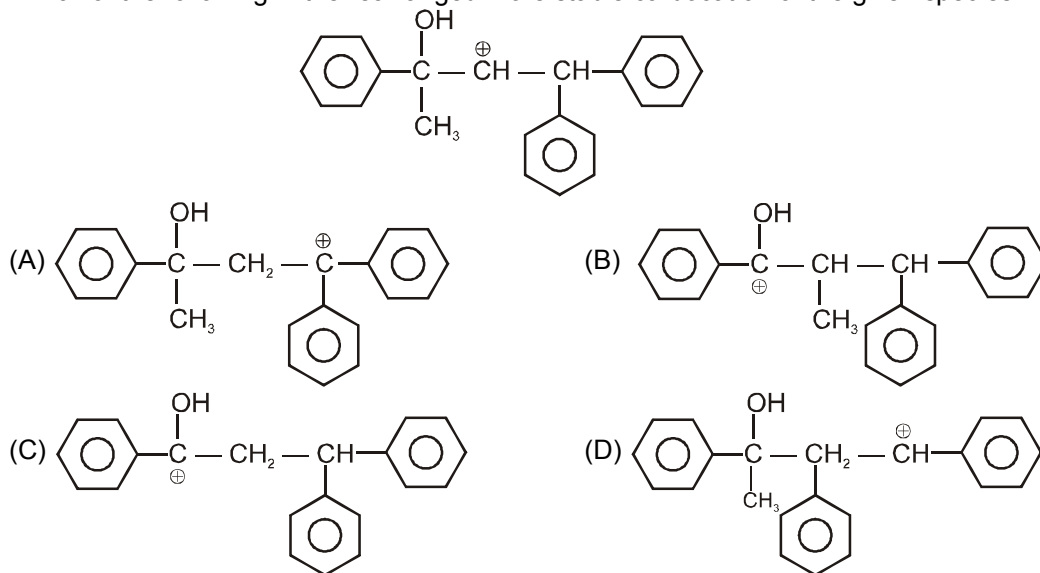




C-5. Most stable rearranged form of given carbocations is :

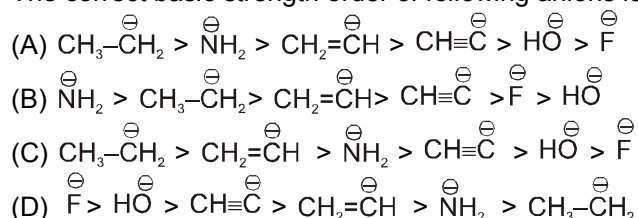


C-6. Which of the following in the rearranged more stable carbocation of the given species?

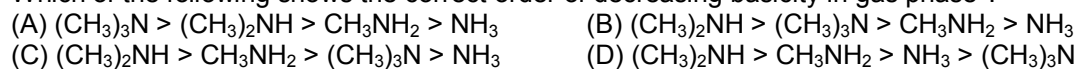


Section (D) : Basic strength

D-1. The correct basic strength order of following anions is :



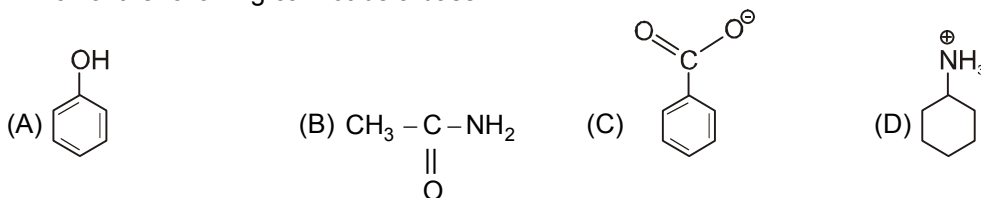
D-2. Which of the following shows the correct order of decreasing basicity in gas phase ?



D-3. Find the order of basic strength. (If R = Me) ?

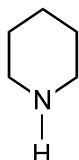


D-4. Which of the following cannot be a base?

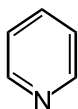




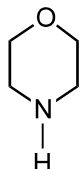
D-5. Select the basic strength order of following molecules ?



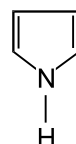
(I) Piperidine



(II) Pyridine



(III) Morpholine

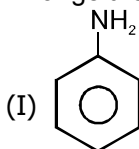


(IV) Pyrrole

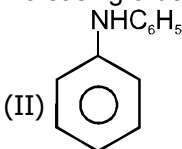
- (A) (IV) > (I) > (III) > (IV)
(C) (II) > (I) > (III) > (IV)

- (B) (III) > (I) > (IV) > (II)
(D) (I) > (III) > (II) > (IV)

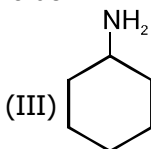
D-6. Arrange the following in increasing order of pKa value ?



(I)



(II)



(III)

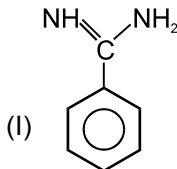
- (A) II < I < III

- (B) III < I < II

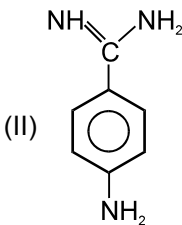
- (C) III < II < I

- (D) II < III < I

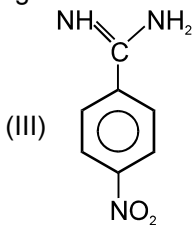
D-7. Select the decreasing order of relative basic strengths of following species :



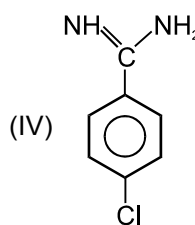
(I)



(II)



(III)



(IV)

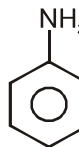
- (A) II > IV > I > III

- (B) III > I > IV > II

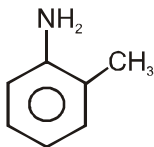
- (C) III > IV > I > II

- (D) II > I > IV > III

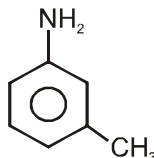
D-8. Select the basic strength order of following molecule :



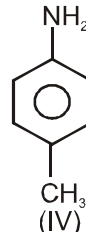
(I)



(II)



(III)



(IV)

- (A) II > III > IV > I

- (B) II > IV > III > I

- (C) IV > II > III > I

- (D) IV > III > I > II

Section (E) : Acidic strength

E-1. Among the following compounds, the strongest acid is :

- (A) $\text{HC} \equiv \text{CH}$

- (B) C_6H_6

- (C) C_2H_6

- (D) CH_3OH

E-2. Which of the following is not correct decreasing K_a order.

- (A) $\text{CH}_4 > \text{NH}_3 > \text{H}_2\text{O} > \text{HF}$

- (B) $\text{CH}_3\text{-OH} > \text{CH}_3\text{-NH}_2 > \text{CH}_3\text{-F} > \text{CH}_3\text{-CH}_3$

- (C) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$

- (D) $\text{PhOH} > \text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH} > \text{CH}_3\text{-C}\equiv\text{CH}$

E-3. Which of the following acid has the smallest dissociation constant ?

- (A) $\text{CH}_3\text{-CH(COOH)-NO}_2$

- (B) $\text{O}_2\text{N-CH}_2\text{-CH}_2\text{-COOH}$



- (C) $\text{Cl-CH}_2\text{-CH}_2\text{-COOH}$

- (D) $\text{NC-CH}_2\text{-CH}_2\text{-COOH}$

E-4. Find the strongest acid among the following compounds is :

- (A) $\text{HOOC-(CH}_2)_2\text{-COOH}$

- (B) $\text{H}_3\text{N}^+ \text{-(CH}_2)_2\text{-COOH}$

- (C) $\text{F-(CH}_2)_2\text{-COOH}$

- (D) $\text{CH}_3\text{-(CH}_2)_2\text{-COOH}$

E-5. Which of the following option shows the correct order of decreasing acidity :

- (A) $\text{PhCO}_2\text{H} > \text{PhSO}_3\text{H} > \text{PhCH}_2\text{OH} > \text{PhOH}$

- (B) $\text{PhSO}_3\text{H} > \text{PhOH} > \text{PhCH}_2\text{OH} > \text{PhCO}_2\text{H}$

- (C) $\text{PhCO}_2\text{H} > \text{PhOH} > \text{PhCH}_2\text{OH} > \text{PhSO}_3\text{H}$

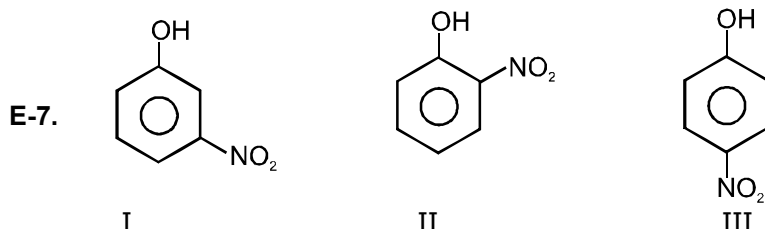
- (D) $\text{PhSO}_3\text{H} > \text{PhCO}_2\text{H} > \text{PhOH} > \text{PhCH}_2\text{OH}$





E-6. Arrange increasing order of acidic strength of following dibasic acids :

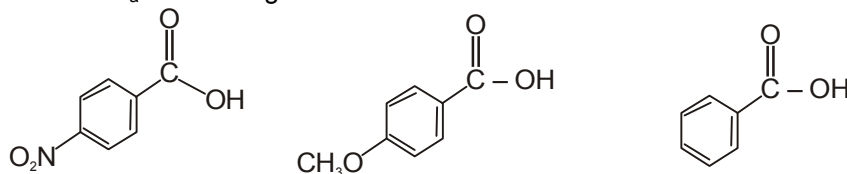
- (I) oxalic acid, (II) succinic acid, (III) malonic acid, (IV) adipic acid
 (A) III < II < I < IV (B) II < III > I > IV (C) I > III > II > IV (D) II > I > III < IV



Arrange above phenol in increasing order of pK_a value :

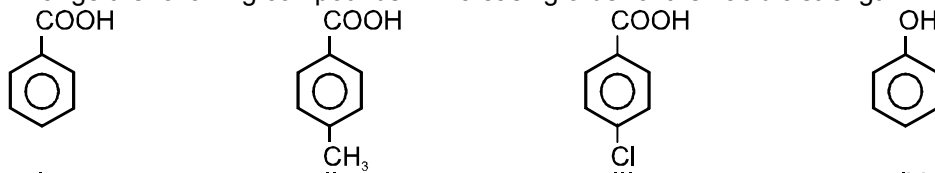
- (A) I < II < III (B) III < I < II (C) III < II < I (D) I < III < II

E-8. Order of K_a of following acids is :



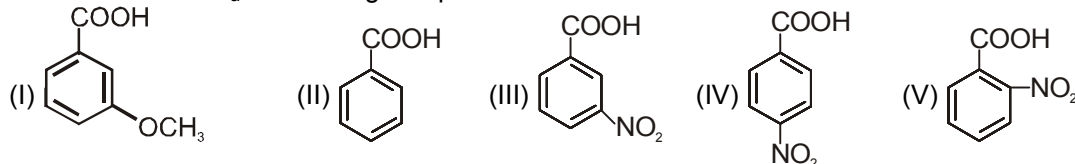
- (A) I > II > III (B) II > I > III (C) I > III > II (D) III > I > II

E-9. Arrange the following compounds in increasing order of their acidic strength.



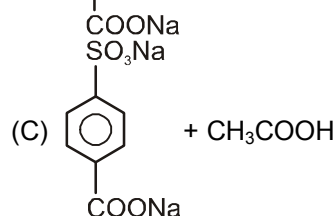
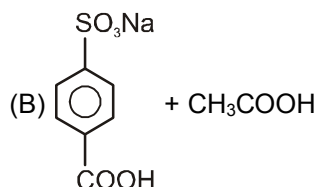
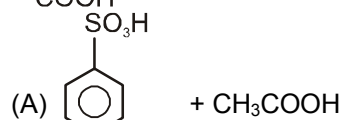
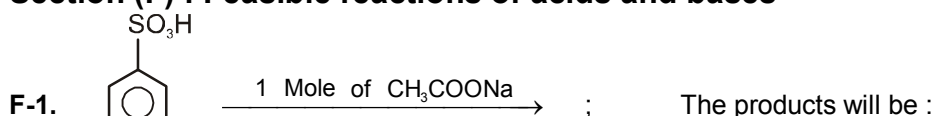
- (A) IV < II < I < III (B) I < II < III < IV (C) IV < II < III < I (D) I < III < II < IV

E-10. Find the order of K_a of following compounds :



- (A) I < II < III < IV < V (B) IV < I < III < II < V (C) III < II < I < IV < V (D) II < I < III < IV < V

Section (F) : Feasible reactions of acids and bases

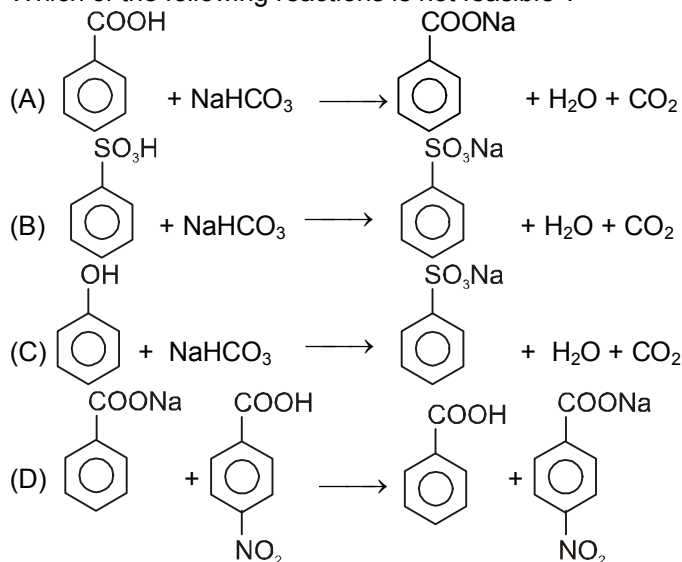


(D) Reaction is not feasible

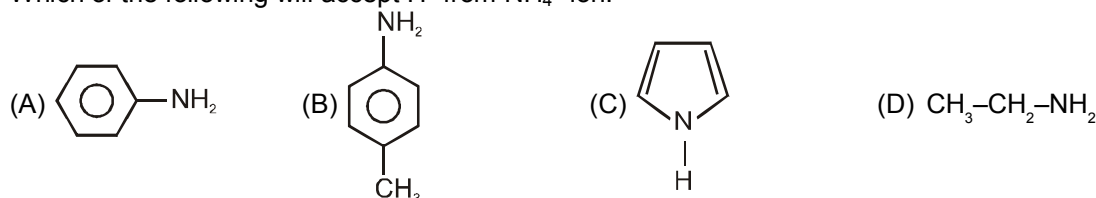




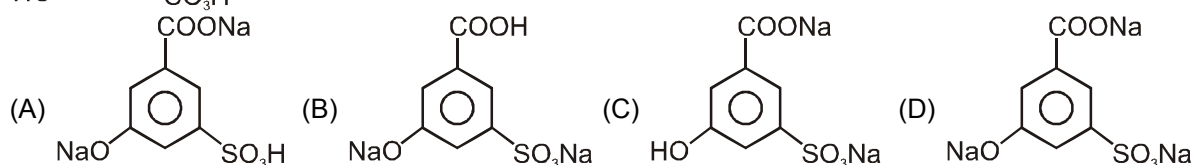
F-2. Which of the following reactions is not feasible ?



F-3. Which of the following will accept H^+ from NH_4^+ ion.

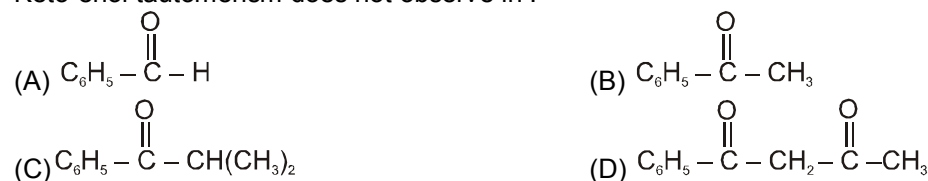


F-4. $\xrightarrow[2 \text{ Mole NaHCO}_3]{\text{Product}}$



Section (G) : Tautomerism

G-1. Keto-enol tautomerism does not observe in :



G-2. The enolic form of acetone contains :

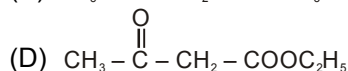
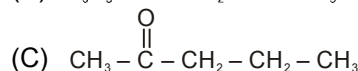
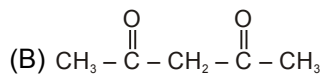
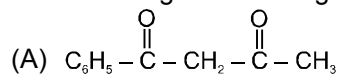
- (A) 9 σ bonds, 1 π bond and 2 lone pairs (B) 8 σ bond, 2 π bond and 2 lone pairs
- (C) 10 σ bond, 1 π bond and 1 lone pair (D) 9 σ bond, 2 π bond and 1 lone pair

G-3. Molecule can be enolised by which hydrogen ?

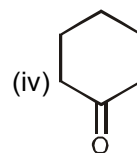
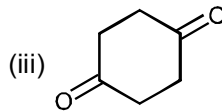
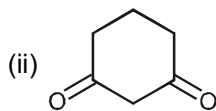
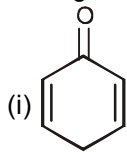
- (A) $y\text{-H}$ (B) $z\text{-H}$ (C) both (D) None of these



G-4. Which among the following compound will give maximum enol content in solution :



G-5. Arrange the following in decreasing order of percentage enol content.



(A) I > II > III > IV

(B) II > I > III > IV

(C) II > III > I > IV

(D) III > II > IV > I

PART - III : MATCH THE COLUMN

1. Match the column :

	Column-I (Keto)		Column-II (% enol)
(A)	$\text{CH}_3 - \text{CH} = \text{O}$	(x)	95 %
(B)	$\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{Ph}$	(y)	76 %
(C)	$\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{OEt}$	(z)	0.0001 %
(D)	$\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$	(w)	7.2 %

2. Match the column :

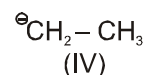
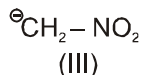
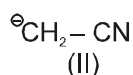
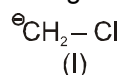
	Column-I		Column-II
(A)	NaHCO_3 will react with	(p)	
(B)	Na will react with	(q)	
(C)	NaOH will react with	(r)	
(D)	NaNH_2 will react with	(s)	

Exercise-2

Marked Questions may have for Revision Questions.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. Arrange the following carbanions in decreasing order of stability :



(A) III > II > IV > I

(B) III > II > I > IV

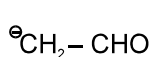
(C) III > I > II > IV

(D) II > III > I > IV

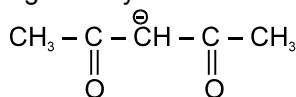




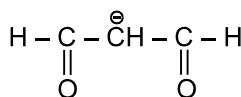
2. Correct decreasing stability order of following carbanions :



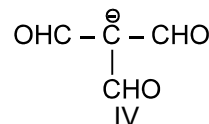
I



II



III



IV

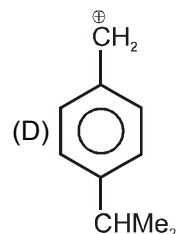
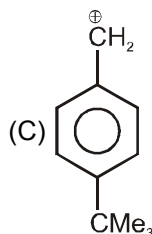
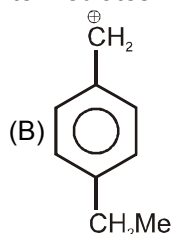
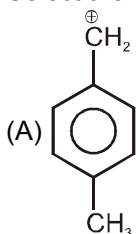
(A) III > IV > II > I

(B) IV > II > III > I

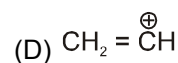
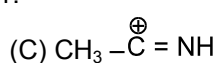
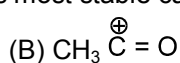
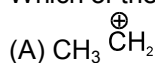
(C) IV > III > II > I

(D) III > II > I > IV

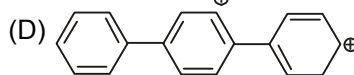
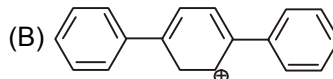
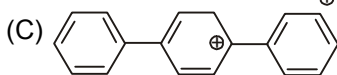
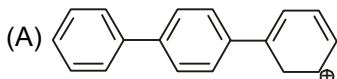
3. Select the most stable intermediates :



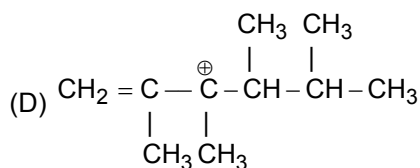
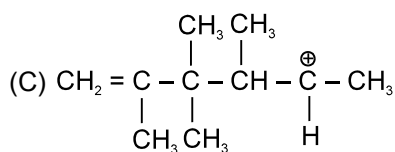
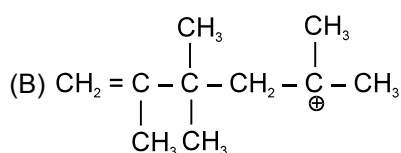
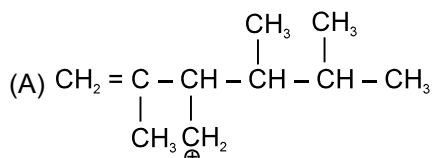
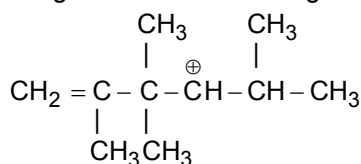
4. Which of the following is most stable carbocation?



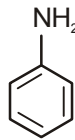
5. The most stable carbocation is :



6. The following carbocation rearranges to

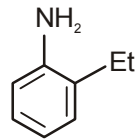


7. Correct basic strength order is :



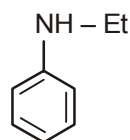
p

(A) r > q > p > s



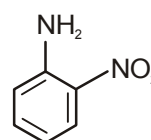
q

(B) r > p > q > s



r

(C) q > r > p > s



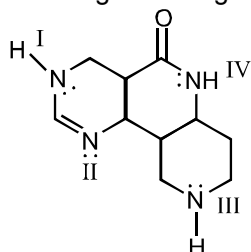
s

(D) r > q > s > p

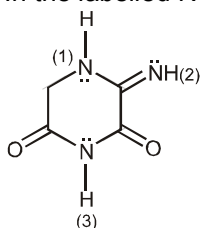




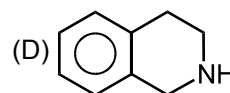
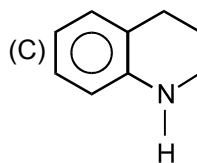
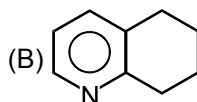
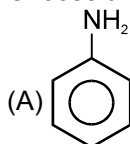
8. The order of basic strength of the given basic nitrogen atoms is :



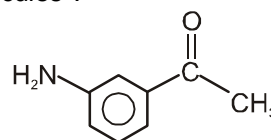
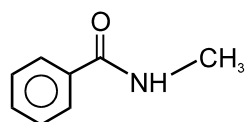
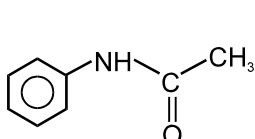
- (A) III > II > I > IV (B) III > I > II > IV (C) I > III > II > IV (D) II > III > I > IV
9. In the labelled N-atoms which is correct basic strength order :



- (A) 2 > 1 > 3 (B) 3 > 1 > 2 (C) 2 > 3 > 1 (D) All are equally basic
10. Choose the strongest base among the following :

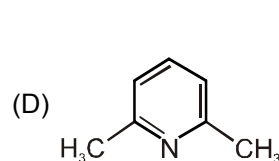
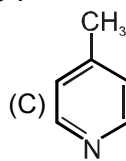
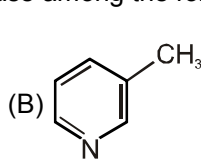
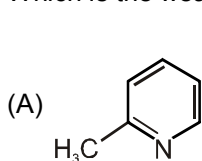


11. Select the basic strength order of following molecules ?

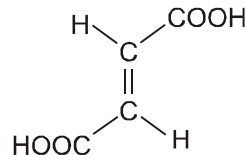
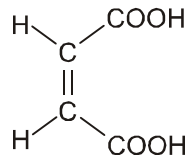
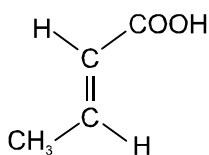


- (A) III > II > I (B) II > III > I (C) I > III > II (D) III > I > II

12. Which is the weakest base among the followings ?



13. Write the order of K_{a1} values of following acids :



- (A) II > III > I (B) I > III > II (C) III > II > I (D) II > I > III

